

Unified Consciousness Theory II

Consciousness Theories as Effective Organization Theories

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Abstract

Major theories of consciousness emphasize different mechanisms (Seth & Bayne, 2022): integrated causal structure, recurrent processing, global availability, higher-order representation, predictive modeling, dendritic integration, memory assembly, attention modeling, and temporospatial organization. Their competition is partly intensified by treating a favored mechanism as a candidate universal boundary between experience and nonexperience.

UCT I (Liu, 2026) adopts a different basal ontology: every actual valid process token possesses experience, structurally identified with a physically anchored, mechanism-fixed intrinsic response structure $\mathfrak{D}^{PA}$. Under that assumption, nonuniversal mechanisms cannot be universal experience-existence gates. UCT II asks whether their *operational cores* can instead be reconstructed, embedded, reinterpreted, or shown to conflict as effective organization theories.

We introduce an admissible translation framework, explicit bridge-admissibility conditions, a two-axis classification separating formal recovery from empirical validation, and anti-triviality criteria requiring translators to be constructive, non-circular, structurally invariant, reusable, failure-prone, and explanatorily nontrivial. A Transla-

tion Fidelity Requirement additionally preserves each target theory’s distinctive signature, strongest phenomenal/ontological claim, and unrecovered empirical residue. We then fidelity-audit nine theories: IIT 4.0 (Albantakis et al., 2023), recurrent processing theory, GNWT, higher-order theories, predictive processing/active inference, dendritic integration theory, memory theory of consciousness, attention schema theory, and temporospatial theory. The result is not a universal endorsement: theory subclaims receive different relation types, formal classes, and validation statuses, including direct conflicts.

Two fixed-parameter toy models provide quantitative cross-theory proof of concept using one fixed parameterization across their theory-linked outputs: a DIT→RPT→GNWT chain and an independent MToC→HOT→GNWT chain. Negative controls dissociate upstream primitives from downstream recurrent, higher-order, and global-access organization. These results establish shared-law reuse only at formal-conditional / toy-model level (F3/V1); they do not constitute held-out human-data recovery (V3), replicated empirical recovery (V4), or robust full-theory compression.

The proposed unification is therefore partial, fidelity-audited, and failure-prone by design. Its governing distinction is that mechanistic recovery, ontological endorsement, and empirical confirmation are separate operations.

Keywords: consciousness; theory unification; intertheory translation; causal organization; IIT; GNWT; higher-order theory; recurrent processing; predictive processing; memory theory; temporospatial theory

1. Why intertheory unification needs constraints

A sufficiently rich mother state can be mapped to arbitrary target labels by lookup.

Therefore:

existence of a map $\neq$ theoretical unification.

A legitimate UCT translator must satisfy:

1. constructive shared basis;
2. non-circular bridges;

3. structural invariance;
4. cross-instance reuse;
5. failure possibility;
6. nontrivial shared reuse or explanatory constraint beyond lookup.

These conditions are necessary but not sufficient. Criterion 6 requires at least nontrivial shared reuse or explanatory constraint; robust whole-theory description-length compression is a stronger optional test, not a prerequisite. A translator can still satisfy the basic constraints while translating only a weakened generic mechanism rather than the actual target theory.

Therefore Paper II adds a **Translation Fidelity Requirement (TFR)**.

A faithful translation must also preserve:

1. source fidelity to the target theory’s published formulation;
2. a minimal distinctive signature Σ_T ;
3. the target theory’s strongest central phenomenal/ontological claim, even when UCT rejects it;
4. target-specific empirical predictions as explicit residue;
5. the distinction between exact primitive recovery and full theory recovery;
6. explicit labeling of any UCT reinterpretation;
7. the possibility that theory fidelity fails even when a generic UCT functional exists.

Thus:

$$\boxed{\text{nontrivial mapping} \neq \text{faithful theory recovery.}}$$

2. Source formalism

UCT I (Liu, 2026) defines:

$$\mathcal{D}^{PA} = (P, s_t, \Theta, C^{PA}, \bar{J}^{MF}, R_\Theta, \mathcal{R}).$$

Every theory-specific translation must label each operation as:

1. MF constitutive query;
2. derived mathematical operation;
3. diagnostic mechanism intervention;
4. semantic/empirical bridge.

This prevents mechanism-changing manipulations from being silently reinterpreted as properties of the current experiential token.

3. Shared operator language

A compact candidate toolkit is:

Res, DoResp, Diff, Comp, Struct, Opt.
--

These denote:

- restriction/coarse-graining;
- intervention-response extraction;
- response comparison;
- composition;
- structural graph/partition/invariant operations;
- explicitly declared selection/optimization rules.

The toolkit is not claimed mathematically minimal.

4. Bridge admissibility and mechanism representation

4.1 Bridge Admissibility Conditions

A theory-specific bridge package B_T is **admissible** only if the following conditions are satisfied.

BAC-1 — Source anchoring

Theory-specific primitives in B_T must be motivated by canonical target-theory sources or by independently identified/measured system structure.

BAC-2 — Output blindness / non-oracularity

B_T may not contain the desired target output, its instance label, or an equivalent lookup oracle for the case being translated. In particular, the bridge cannot count as a recovery device merely because it encodes the answer that $\mathcal{T}_T$ is supposed to reconstruct.

BAC-3 — Cross-instance fixation

For a declared domain D_T , the bridge rules are fixed independently of the particular target result. Instance-specific measured variables may vary; the bridge definition itself may not be redesigned for each case.

BAC-4 — No post-hoc target refitting

Translator structure, thresholds, and bridge semantics must not be changed after observing the target-theory output for each evaluated instance.

BAC-5 — Structural covariance

When both UCT and the target observable are invariant under an admissible structure-preserving reparameterization, the translated result must be invariant as well.

BAC-6 — Failure possibility

There must exist possible systems or cases for which the translation can fail source fidelity, fail a target prediction, or remain undefined because the required bridge conditions are absent.

BAC-7 — Residual transparency

Anything imported from the target theory rather than generated by the shared UCT formalism must remain explicit in B_T or in the uncovered residual set R_T^{res} .

These conditions formalize the anti-lookup and anti-oracle requirements used throughout Paper II.

4.2 Admissible Mechanism Representation Proposition

Call a mechanism theory T admissible over domain D_T if each operational observable can be written:

$$O_j = E_j(\mathfrak{D}^{PA}, B_T)$$

using the shared language and a **BAC-admissible**, fixed bridge package B_T .

Then a compositional translator exists:

$$\mathcal{T}_T : (\mathcal{D}^{PA}, B_T) \mapsto O_T.$$

The proof is structural induction over the declared expression tree.

This proposition establishes compositional closure of the admissible translation language. It does **not** by itself establish that a proposed B_T satisfies BAC, that every consciousness theory belongs to the class, that the target theory is empirically correct, or that the translated theory has been fully reduced.

5. Two-axis classification and effective organization descriptions

The earlier single R0-R5 ladder mixed formal relation/recovery status with empirical validation. Paper II therefore uses two independent axes.

5.1 Formal relation/recovery class F

Class	Meaning
F0	direct conflict for the specified subclaim
F1	logical compatibility only
F2	embedding / representability
F3	conditional derivation under an explicit BAC-admissible bridge
F4	exact/local formal reduction of the specified primitive

These classes apply to **subclaims or primitives**, not whole theories. They are a formal classification vocabulary, not a total-order score over whole theories; different subclaims of the same theory may occupy different classes. They are not empirical validation scores.

5.2 Empirical validation class V

Class	Meaning
V0	no empirical validation claim
V1	toy-model / model-organism proof of concept

Class	Meaning
V2	fitted or in-sample empirical recovery
V3	preregistered or otherwise fixed held-out recovery against relevant baselines
V4	replicated / cross-system empirical effective recovery

The two axes are independent. In particular:

$$F4 \not\Rightarrow V3$$

and:

$$V3 \not\Rightarrow F4.$$

An exact graph-theoretic primitive may be $F4/V0$, while an empirically successful effective model could be $F2$ or $F3$ with $V3$ or $V4$.

5.3 Relation labels and residuals

For each target subclaim, the UCT relation label is drawn from:

- **REC** — formal reconstruction;
- **EMB** — embedding;
- **REI** — UCT reinterpretation;
- **CON** — direct conflict.

Empirical/unrecovered material is **not** a fifth relation type. It is recorded separately in:

$$R_T^{res}$$

and in the empirical validation class V_T .

This separation prevents logical relation, residual content, and evidential status from being conflated.

5.4 Effective organization description

A target-theory subclaim c is an **effective organization description relative to UCT** when:

1. c concerns a nonuniversal organization property rather than the universal existence predicate for experience;
2. its operational content is reconstructible or embeddable from $(\mathfrak{D}^{PA}, B_T)$ under BAC;
3. any target-specific phenomenal or ontological interpretation is separately labeled rather than silently inherited;
4. unrecovered commitments remain explicit in R_c^{res} .

Thus:

$$EOD_{UCT}(c) \Rightarrow \text{organization recovery without automatic ontological endorsement.}$$

This is the sense in which Paper II treats recoverable parts of consciousness theories as **effective organization theories**.

5.5 Scope and nonuniversality-witness rule

Any use of the Organization-Gate Impossibility result or any F0/CON classification must state the relevant domain and the condition that makes the target mechanism nonuniversal.

For a candidate organization predicate $M(P)$, “nonuniversal” means that there exists at least one actual valid UCT token in the declared domain for which the predicate is absent or falls outside the target region:

$$\exists Q_{\text{actual}} \in D : \neg M(Q).$$

Only then does A1 yield the gate contradiction:

$$E(Q) = 1 \quad \text{while} \quad M(Q) = 0.$$

Likewise, a direct conflict classification must identify the exact target-theory necessity/identity claim being contradicted, rather than infer conflict from a theory label alone. This rule makes domain and existence witnesses explicit rather than leaving them as hidden premises.

5.6 Dependency separation: reconstruction, no-gate relocation, and experiential interpretation

Paper II distinguishes three logically different operations.

Formal reconstruction

Whether a target operational observable can be reconstructed or embedded from:

$$(\mathfrak{D}^{PA}, B_T)$$

under BAC is a formal/mechanistic question. The existence of such a translator does not, by itself, empirically prove A1 or A5.

No-universal-gate relocation

For a target mechanism that is nonuniversal on the declared UCT domain, A1 plus the required nonuniversality witness imply that the mechanism cannot equal the universal experience-existence predicate.

Experiential-organization interpretation

Treating a BAC-admissibly recovered physical organization as part of **experiential organization** additionally depends on A5:

$$\Phi(P) \equiv_{\text{struct}} \mathfrak{D}^{PA}(P).$$

Thus:

formal reconstruction $\neq$ no-gate relocation $\neq$ experiential-organization interpretation.

The empirical validation axis V is separate from all three.

6. IIT 4.0

Given:

- an IIT-compatible chart;
- current state;
- mechanism-fixed transition/intervention data;
- IIT's prior, intrinsic-difference metric, partition family, max/min rules and tie conventions;

IIT computational quantities are conditionally reconstructible once IIT's own formal conventions are supplied.

Thus:

$$IIT_{\text{computational}} : F3/V0.$$

If, under IIT's own units, repertoire definitions, prior, intrinsic-difference metric, and admissible partition rules, the IIT-defined cause/effect repertoires factorize so that the partitioned and unpartitioned repertoires have zero intrinsic difference, then the corresponding IIT mechanism-level integrated information is:

$$\phi_s = 0.$$

Generic graph factorization alone is not claimed to be sufficient for this IIT result.

However UCT does not inherit maximal exclusion.

$$IIT_{\text{exclusion ontology}} : F0$$

because, when the lower process remains an actual valid UCT token, A3 does not permit overlap or nonmaximality **alone** to erase that lower process's experience.

This is a model of formal reconstruction plus explicit reinterpretation/conflict, not a full reduction of IIT as a phenomenal theory.

7. Recurrent Processing Theory

From the intervention-defined directed causal graph $G_C(P)$, strongly connected components identify recurrent causal modules.

This yields exact recovery only of a **primitive graph-recurrence property**:

$$Primitive_{\text{graph recurrence}} : F4_{\text{local}}/V0.$$

A faithful RPT translation additionally requires a sensory/perceptual chart, content-sensitive recurrence, theory-relevant timing, and a

stabilization/perceptual consequence. Therefore the theory-level status remains:

$$RPT_{\text{theory}} : F3/V0.$$

But the claim that a particular recurrent neural regime is necessary/sufficient for perceptual consciousness remains a theory-specific empirical bridge:

$$RPT_{\text{phenomenology}} : \text{residual empirical claim; current } V0.$$

8. Global Neuronal Workspace

For content process S_c and declared consumers K_j , define causal access:

$$a_j(c, t; \tau) = d_j[P(X_{t+\tau}^{K_j} \mid do(C_c = c)), P(X_{t+\tau}^{K_j} \mid do(C_c = c_0))].$$

This yields an access vector, breadth/coverage measures, persistence, and workspace macrostate.

Thus:

$$GNWT_{\text{core}} : F3/V0.$$

UCT does not equate workspace ignition with experience existence.

Strong GNWT anatomical/timing predictions remain empirical and may fail independently of the existence of a generic access functional.

9. Higher-Order Theory and Attention Schema Theory

A higher-order process may causally track a lower-order state.

AST further specializes this into a simplified internal model of attention used to control future attention.

These causal organizations are representable.

But:

causal targeting $\neq$ fully grounded semantic aboutness.

Therefore:

$Primitive_{\text{hierarchical causal tracking}} : F4_{\text{local}}/V0, \quad HOT_{\text{theory}} : F3/V0,$

and:

$$AST_{\text{mechanistic}} : F3/V0.$$

A unified representation-grounding bridge remains required.

10. Predictive Processing and Active Inference

A PP/AI chart requires extra structure:

- agent/environment partition;
- internal model state;
- latent variables;
- p_{θ} ;
- q_{ϕ} ;
- prediction/error relations;
- policies;
- preferences/objectives.

Once those bridges are fixed, free-energy and policy quantities are computable functionals of the process chart.

Thus:

$$PP/AI : F3_{\text{conditional}}/V0.$$

UCT does not infer generative-model semantics from bare topology alone.

Furthermore, PP/AI recovery requires independently motivated model-state identification and comparison against plausible non-PP/AI computational alternatives. A generative-model fit that can be redefined per instance does not satisfy TFR.

11. Dendritic Integration Theory

Specify biologically identified:

- apical input/state;
- basal input/state;
- somatic output;
- thalamic modulation.

Nonlinear interventional interaction and downstream propagation become response functionals of the physical process.

However, nonlinear gating alone is not DIT. A faithful DIT signature must preserve:

- pyramidal-cell compartment structure;
- bottom-up / top-down integration;
- apical/basal interaction;
- thalamocortical modulation/propagation;
- the biological conscious-state claims associated with the theory.

Thus a local interaction functional is only a primitive. The theory-level relation remains:

$$DIT_{\text{theory}} : F3/V0.$$

Dendritic gating can affect later recurrence and global propagation without being the universal origin of experience.

12. Memory Theory of Consciousness

A minimal temporal-retention relation can be defined by whether changing an earlier state changes a later independently specified retained/trace-candidate state.

This gives:

$$Primitive_{\text{temporal causal retention}} : F4_{\text{local}}/V0.$$

Identifying working, episodic, semantic, and assembly systems requires empirical bridges:

$$MToC_{\text{mechanistic}} : F3/V0.$$

On the strong necessity reading preserved in the source-trace ledger, explicit memory is not the universal experience-existence gate **provided that the declared UCT domain contains at least one actual valid process token lacking the target explicit-memory system.**

Let $M_{\text{explicit}}(P)$ denote possession of that target system. If:

$$\exists Q_{\text{actual}} : \neg M_{\text{explicit}}(Q),$$

then A1 gives $E(Q) = 1$, whereas a universal explicit-memory necessity claim gives $E(Q) = 0$. Under this explicit nonuniversality witness, the strong identity is an F0/CON conflict:

$$\boxed{MToC_{\text{all-experience via explicit-memory}} : F0.}$$

Thus temporal causal retention is an exactly recoverable structural primitive, identifying that retained state as a memory system remains bridge-dependent at F3/V0, and the strongest universal MToC identity remains visible as a rejected claim.

13. Temporospatial Theory

UCT is already indexed by time horizon and scale:

$$\mathcal{D}_{t,\tau,g}^{PA}.$$

Nestedness, alignment, expansion, and globalization can therefore be represented by cross-scale causal relations, state-input interaction, causal support growth, and global reachability.

Thus:

$$TTC_{\text{structural}} : F3/V0$$

with several local direct functionals.

The mapping from these quantities to phenomenology remains an empirical/theoretical bridge.

14. Fidelity-audited nine-theory matrix

14.1 Distinctive signatures retained

Theory	Distinctive signature retained
IIT	IIT priors/ID metric, partitions, max-min rules, and composition
RPT	sensory/content-sensitive recurrence plus timing and stabilization
GNWT	broad consumer access, amplification/persistence, and ignition
HOT	higher-order representation, not mere causal tracking
PP/AI	fixed generative, inference, and policy architecture
DIT	dendritic, pyramidal, and thalamocortical signature
MToC	explicit-memory architecture plus the strong all-experience claim
AST	internal attention model plus control role
TTC	nestedness, alignment, expansion, and globalization

14.2 Formal relation and validation matrix

Theory	Formal status	Relation labels	Validation
IIT	computational F3; exclusion F0 under scoped overlap witness	REC + REI + CON	V0

Theory	Formal status	Relation labels	Validation
RPT	graph recurrence F4; theory F3	REC + EMB + REI	V0
GNWT	core F3	EMB + REI	V0
HOT	causal- tracking primitive F4; theory F3	REC + EMB + REI	V0
PP/AI	F3 conditional	EMB + REI	V0
DIT	local interaction F4; theory F3	REC + EMB + REI	V0
MToC	temporal retention F4; theory F3; strong identity F0 given nonuniversal- ity witness	REC + EMB + CON	V0
AST	F3	EMB + REI	V0
TTC	F3	EMB + REI	V0

14.3 Residual and conflict status

Theory	Main residual/conflict
IIT	exclusion conflict; neural/temporal predictions
RPT	phenomenal/perceptual sufficiency, timing/localization
GNWT	strong neural/temporal implementation
HOT	representational semantics and phenomenal sufficiency
PP/AI	model semantics + comparative validation
DIT	biological necessity/sufficiency and implementation
MToC	strong ontology conflict; timing/anatomy
AST	semantic aboutness, awareness attribution, control predictions
TTC	neurophenomenal mapping and scale/frequency predictions

$$F4_{\text{primitive}} \not\Rightarrow F4_{\text{theory}}$$

and independently:

$$F4 \not\Rightarrow V3.$$

Comparison-only frameworks. Information Closure Theory (ICT) and Integrated World Modeling Theory (IWMT) remain important comparison/unification competitors in Section 15, but they are not counted as members of the nine-theory source-trace atlas audited in Section 14.

15. Nearest unification competitors and novelty boundary

UCT does not claim to be the first framework to integrate, reconcile, decompose, or compare existing consciousness theories.

Important prior art includes:

- hierarchical causal descriptions (Coward & Sun, 2004);
- IWMT as a synthetic framework combining IIT, GNWT, the Free Energy Principle, and active inference (Safron, 2022), with a 2026 extension relating IWMT to the Human Consciousness Hypothesis (Safron et al., 2026);
- Information Closure Theory (Chang et al., 2020);
- TICL (Winters, 2020, 2021);
- universal-theory criteria that seek intrinsic mathematical formulations (Kanai & Fujisawa, 2024);
- the temporal-hierarchy/minimal-model unification of six prominent consciousness theories by Singhal and Srinivasan (2026);
- the building-block approach of Lopez and Wiese (2025), which explicitly recommends shifting from whole theories to simpler elements that can support assessment, integration, and unification;
- Tonetto’s (2026) preprint separating recurring structural findings from broader ontological commitments across major consciousness theories.

These works substantially narrow any defensible novelty claim. Paper II therefore does **not** claim novelty for the general ideas that theories may be complementary, that whole theories can be decomposed into smaller commitments, or that multiple theories can be organized inside a larger synthesis.

The contribution claimed here is the conjunction of four more specific moves:

1. a physically anchored mother formalism inherited from UCT I;
2. explicit **Bridge Admissibility Conditions** plus a **Translation Fidelity Requirement** that preserve non-oracular fixed

bridges, a target-specific signature Σ_T , the strongest target claim, and explicit residual commitments R_T^{res} ;

3. independent relation labels for formal reconstruction, embedding, UCT reinterpretation, and direct conflict, with empirical validation and unrecovered residual commitments tracked on separate axes rather than collapsed into a single global claim that a theory has been “reduced”;
4. failure-prone quantitative shared-law reuse in fixed-parameter toy models with negative controls and a single parameterization shared across the theory-linked outputs.

The UCT-specific methodological rule is therefore:

reconstruct or embed an operational mechanism while independently classifying its ontology

When UCT changes a target theory’s explanatory role—from consciousness generator or universal criterion to organization phenotype—that move is labeled **reinterpretation**, not neutral reduction.

The **no-universal-gate relocation** follows from A1 together with the relevant nonuniversality witness. Interpreting a BAC-admissibly recovered physical mechanism as part of **experiential organization** additionally depends on A5. Paper II therefore does not treat A1 alone as sufficient for the full experiential-organization interpretation.

16. Cross-theory consequences

A mother framework becomes more interesting when it generates constrained relations between theory-linked mechanisms. The arrow-like notation in this section denotes **conditional enabling**, not unconditional sufficiency.

Examples:

DIT-like gating $\rightsquigarrow$ RPT-like recurrence under a return-path condition

Opening a causal edge $u \rightarrow v$ creates a new directed cycle iff a return path $v \rightsquigarrow u$ already exists. Interpreting the gated edge as DIT-relevant and the resulting cycle as RPT-relevant additionally requires the corresponding DIT and RPT bridge conditions.

Recurrence as an enabling condition for ignition-like transitions

Restricted recurrent dynamical systems **can** undergo saddle-node/bistable transitions yielding GNWT-like ignition when the required nonlinear feedback, drive, and workspace conditions are present. Recurrence alone is not asserted to be sufficient:

recurrence $\nRightarrow$ global access.

Grounded representation + broad availability $\rightsquigarrow$ globally available internal model

An independently grounded representation that becomes causally available to a declared broad consumer family yields a globally available internal model relative to that declared access architecture.

These are conditional/enabling cross-theory statements, not unconditional implications and not proofs of the target theories.

17. Empirical program

Paper II can be complete as a formal theoretical preprint at V0/V1; the **empirical-recovery program** is not complete until at least fixed held-out recovery against relevant baselines (V3), with cross-system replication (V4) as the stronger target.

Required tests include:

1. preregistered bridge laws;
2. fixed translator parameters across systems;
3. held-out recovery of target-theory quantitative effects using fixed bridge/translator rules, with only declared measured inputs allowed to vary;
4. baseline comparison;
5. cross-theory predictions not imported from the target theory;
6. explicit failure leading to bridge revision rather than post-hoc relabeling.

The earlier GNWT toy dynamics and the present shared-law examples are V1 model evidence, not V3/V4 empirical recovery. The 2025 adversarial collaboration is especially important because it directly challenged key preregistered predictions of both IIT and GNWT rather than merely comparing verbal frameworks (Cogitate Consortium et al., 2025).

18. Faithfulness as a publication constraint

Paper II should not count a target theory as “recovered” unless a reader who endorses that target theory could recognize its distinctive machinery and its strongest claim in the translated representation.

The following failure mode is now explicitly prohibited:

delete target-specific commitments $\rightarrow$ retain generic mechanism $\rightarrow$ declare unification.

For each theory T , the release package should preserve:

1. source passages/formal definitions establishing Σ_T ;
2. the UCT translator;
3. the relation labels REC/EMB/REI/CON, the formal class F , and empirical validation class V ;
4. theory-specific predictions left in R_T^{res} ;
5. conditions under which the translation would fail fidelity.

This is stronger than the original anti-lookup criterion because it prevents **conceptual overcompression** as well as mathematical overfitting.

19. Quantitative shared-law example: DIT $\rightarrow$ RPT $\rightarrow$ GNWT

Paper II’s nontriviality criterion requires more than writing compatible translation formulas. At least one shared dynamical law should generate observables associated with more than one target-theory signature under one fixed parameterization shared across the linked outputs.

A minimal fixed model was therefore constructed with one scalar gate parameter:

$$g \in [0, 1],$$

mapped to:

$$q(g) = \sigma[10(g - 0.5)].$$

The same gate enters all three theory-linked observables.

19.1 DIT-like local interaction

For apical-like input a and basal-like input b ,

$$d(a, b; g) = \sigma[-2 + 1.2a + 1.2b + 3q(g)ab].$$

Define:

$$\Delta_{AB} = d(1, 1) - d(1, 0) - d(0, 1) + d(0, 0).$$

This gives a smooth DIT-like nonlinear interaction measure.

19.2 RPT-like recurrent gain

A two-node recurrent core contains one fixed return edge $r = 2$ and one gated edge $3q(g)$.

The structural spectral gain is:

$$\rho(g) = \sqrt{6q(g)}.$$

Therefore:

$$\boxed{\rho = 1 \quad \text{at} \quad g \approx 0.339.}$$

19.3 GNWT-like access

The same gated edge enters a fixed nonlinear workspace:

$$x_1 = \sigma(5[-0.5 + s + 2x_2]),$$

$$x_2 = \sigma(5[-0.5 + 3q(g)x_1]).$$

Five fixed downstream consumers define global availability G .

With baseline content drive $s = 0.3$, the high-access operational criterion

$$G \geq 0.8$$

is first reached at approximately:

$$g \approx 0.403.$$

The reported model uses the same shared gate parameter and fixed GNWT readout architecture across the DIT-, RPT-, and GNWT-linked outputs; no separate gate parameter is introduced for the GNWT readout.

19.4 Negative control

The local DIT-like interaction is left unchanged, but the return path is removed.

Then:

$$\rho = 0$$

for the recurrent core and the maximum global-access score remains approximately:

$$G_{\max} \approx 0.0397.$$

Thus:

$$\text{local gating alone} \not\Rightarrow \text{recurrent criticality} \not\Rightarrow \text{global access.}$$

19.5 Unrefitted drive perturbation sweep

All model parameters are held fixed while only content drive is changed. This is a **model-level unrefitted perturbation sweep**, not empirical held-out validation in the V3 sense.

The recurrent structural threshold remains fixed at $g \approx 0.339$, whereas GNWT-like high access shifts with drive and disappears completely for sufficiently weak drive.

For example:

- $s = -0.4$: no high-access transition over the full gate sweep;
- $s = -0.3$: high access appears near $g = 0.575$;
- $s = -0.2$: near $g = 0.475$;
- $s \geq 0$: near $g \approx 0.403$.

This supplies a useful dissociation:

recurrent gain above threshold is not sufficient for global access.

19.6 Status of the result

This is not biological validation of DIT, RPT, or GNWT.

Its significance is methodological:

- one model;
- one shared gate parameter;
- three theory-linked observables;
- a negative control;
- unfitted drive perturbation sweep;
- one fixed parameterization shared across the theory-linked outputs.

It therefore counts as:

shared quantitative-law reuse at $F3/V1$ level.

20. Second quantitative shared-law example: MToC $\rightarrow$ HOT $\rightarrow$ GNWT

To avoid relying on one biological/network chain, a second independent model was constructed around temporal retention, higher-order tracking, and global access.

A single content pulse initializes:

$$m_0 = 1.$$

Memory retention is controlled by one parameter:

$$m_{t+1} = \lambda m_t.$$

A fixed higher-order tracking state obeys:

$$h_{t+1} = 0.6h_t + 0.4\sigma[12(m_t - 0.35)].$$

A fixed recurrent workspace obeys:

$$w_{t+1} = 0.7w_t + 0.3\sigma[10(h_t + 1.2w_t - 0.8)].$$

Five fixed consumer channels define the global-access score G .

The same retention parameter λ is used across the three fixed read-outs; no separate output-specific retention parameter is introduced.

20.1 Distinct transitions

MToC-like retention varies continuously as:

$$m_5 = \lambda^5.$$

A primitive higher-order activation criterion:

$$\max_t h_t \geq 0.5$$

is first reached at approximately:

$$\lambda_{\text{HOT}} \approx 0.403.$$

GNWT-like high access:

$$\max_t G_t \geq 0.8$$

is first reached at approximately:

$$\lambda_{\text{GNWT}} \approx 0.722.$$

Thus:

$$\text{retention} \rightarrow \text{higher-order activation} \rightarrow \text{global access}$$

occurs in one fixed model, but the three organization coordinates are not identical.

20.2 Edge-cut controls

If the memory-to-higher-order edge is cut:

- the memory trace remains;
- higher-order activation collapses;
- GNWT-like high access disappears.

If the higher-order-to-workspace edge is cut:

- memory remains;
- higher-order activation remains;
- GNWT-like high access disappears.

Hence:

retention alone $\Rightarrow$ higher-order organization $\Rightarrow$ global availability.

20.3 Fidelity boundary

This model does **not** reconstruct full MToC, HOT, or GNWT.

- For MToC it models a retention/temporal-assembly primitive, not the strong claim that the explicit-memory system is responsible for all conscious experience.
- For HOT it models a higher-order tracking primitive, not complete semantic representation.
- For GNWT it models a recurrent global-access transition, not its full neural implementation.

Accordingly:

$$F4_{\text{primitive}} \not\Rightarrow F4_{\text{theory}}.$$

The significance is methodological: this is a second independent example in which one fixed parameterization produces observables associated with multiple target-theory signatures while preserving clear failure modes.

21. Source-trace and compression status

The Translation Fidelity Requirement remains source-grounded in the nine-theory v0.54 source-trace ledger, while rc4 adds a superseding formal-relation ledger and dependency map that separate source

evidence, formal class, relation type, empirical validation, residual commitments, and explicit fidelity-failure conditions.

For each theory T , Paper II records an audit tuple:

$$\mathcal{A}_T = (D_T, \Sigma_T, B_T, \mathcal{T}_T, R_T^{res}, F_T, \Lambda_T, V_T),$$

where D_T is the declared target domain, Σ_T the minimal distinctive signature, B_T the BAC-admissible bridge package, $\mathcal{T}_T$ the translator, R_T^{res} the unrecovered commitments, F_T the explicit fidelity-failure conditions, Λ_T the relation labels, and V_T the empirical validation status.

A separate symbolic description-length proxy was used to test whether the shared UCT formalism actually compresses the nine theory descriptions. Six shared operator classes are reused 44 times, so shared formal reuse is real. However the net compression gain changes sign under conservative assumptions about shared-core and bridge cost.

Therefore the compression claim remains restricted to **shared formal reuse**, not demonstrated robust explanatory compression.

The required quantitative-reuse step has now been met at proof-of-concept level by the two fixed-parameter shared-law models in Sections 19 and 20. The next major scientific burden is preregistered held-out empirical recovery, comparison against relevant baselines, and evidence that any shared representation yields predictive or compression advantages beyond the toy-model setting.

22. Claim-status summary

Paper II claim	Status
A common physically anchored mother formalism can express target-theory primitives	formal framework claim
Arbitrary lookup mappings do not count as unification	methodological criterion
TFR is required in addition to anti-triviality	methodological requirement
IIT formal quantities are conditionally reconstructible	F3 / REC / V0

Paper II claim	Status
IIT exclusion ontology is compatible with UCT	no; F0 / CON under scoped A3
RPT graph-recurrence primitive is exactly definable on a fixed chart	F4 primitive / V0
GNWT global-access core is conditionally embeddable	F3 / EMB / V0
HOT hierarchical causal-tracking primitive is exact	F4 primitive / V0; semantics not included
PP/AI is recoverable without model-semantics bridges	no
Temporal causal retention is exact/local	F4 primitive / V0; memory identity requires bridge
Strong MToC explicit-memory identity is accepted by UCT	no; F0 / CON under A1 plus an explicit nonuniversality witness
Nine-theory source fidelity is fully proven	no; working source-trace audit only
Shared formal reuse exists	yes
Robust full-theory compression is established	no
First DIT→RPT→GNWT shared-law model exists	yes, F3/V1 proof of concept
Second MToC→HOT→GNWT shared-law model exists	yes, F3/V1 proof of concept
Held-out empirical recovery (V3) has been achieved	no
Human-data validation has been achieved	no
UCT II proves the target theories true	no

23. Reproducibility and release materials

The public release package preserves the original nine-theory source-trace ledger for source provenance, the rc4 formal-relation ledger and dependency map for the repaired taxonomy, compression-sensitivity records, both shared-law model specifications, and the archived CSV outputs used for the numerical claims in Sections 19 and 20.

The original model-generation script was not preserved in the located research artifacts. During the v1.0-rc3 prepublication audit, a **release-side reconstruction script** was therefore built from the archived equations and CSV outputs. This reconstruction is explicitly not represented as the original code.

Accordingly, Paper II claims a **fixed-parameter cross-output representation in the preserved specification and release-side reconstruction**. Because the original generation script and full development chronology are unavailable, the release audit does not independently certify the stronger historical claim that no parameter tuning of any kind occurred during the original construction process.

Under the declared five-consumer readout form for the first model, the archived outputs recover the fixed consumer-bias multiset

$$\{-5.2, -5.1, -5.0, -4.9, -4.8\},$$

with consumer gain 7. The same readout reproduces the archived second-model access scores. For the second model, the archived aggregate traces are reproduced by 39 state transitions after $t = 0$, with $h_0 = w_0 = 0$, as documented in the release-side audit script.

Across the six regenerated core CSV files, the reconstructed implementation matches the archived outputs within a maximum absolute numerical discrepancy below 10^{-9} (approximately 6.7×10^{-11} in the least exact table and near floating-point precision in the second model).

This audit supports reproducibility of the **reported toy-model numerical claims**. It does not turn the models into biological validation, does not establish V3/V4 empirical recovery, and does not establish that the reconstruction is byte-identical to the unavailable original generation code.

24. Conclusion

UCT II proposes a constrained form of theory unification.

It does not say that IIT, RPT, GNWT, HOT, PP/AI, DIT, MToC, AST, and TTC are all literally true descriptions of the same phenomenal boundary.

It says that identifiable operational mechanisms from these theories can be represented and classified inside one physically anchored process formalism.

The governing rule is:

mechanistic recovery $\neq$ ontological endorsement $\neq$ empirical confirmation.

If this framework is useful, its strongest future evidence will not be the existence of translation formulas, but quantitative compression, held-out recovery, and genuinely novel cross-theory predictions.

Two independent shared-law toy models now satisfy a minimal quantitative-reuse criterion at F3/V1, but neither constitutes held-out human-data recovery (V3), replicated effective recovery (V4), or robust full-theory compression. The framework is therefore offered as a constrained theoretical architecture and empirical research program, not as a replacement for experimental adjudication among consciousness theories.

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